

Unit III — Graphs & Isomorphism

Bipartite Knowledge Graphs and Permutation-Similarity Certificates

GROUP 4 • Discrete Mathematics • August 28, 2026

Educational mathematics assignment — not medical advice.

1. Case Title (Unit III Focus)

Modeling symptom–disease knowledge as a bipartite graph and proving that two different drawings are the same knowledge base.

2. Problem Statement (Unit III)

Can we model the knowledge base faithfully as a bipartite graph $G=(S \cup D, E)$? How do we convince a reviewer that two drawings G and G' with different layouts are isomorphic — i.e., the same knowledge?

3. Real-world Context

Medical editors maintain symptom–disease edges in different tools. Merging edits requires proving isomorphism, not trusting layout. Bipartite structure also enables 2-hop diagnosis: patient $\rightarrow$ symptom $\rightarrow$ disease.

4. Mathematical Concept Used

Bipartite graphs, degree sequences, biadjacency matrices, permutation matrices, and the certificate $A_2 = P \cdot A_1 \cdot P^T$.

5. Mathematical Formulation

$V = S \cup D$, $S = \{S_1 \text{ Fever}, S_2 \text{ Cough}, S_3 \text{ Rash}, S_4 \text{ Fatigue}\}$, $D = \{D_1 \text{ Influenza}, D_2 \text{ Measles}, D_3 \text{ Anemia}\}$, $|V| = 7$, $E = \{S_1 D_1, S_1 D_2, S_2 D_1, S_3 D_2, S_4 D_1, S_4 D_3\}$, $|E| = 6$. Second representation reorders as $(S_4, S_3, S_2, S_1 \mid D_3, D_2, D_1)$.

6. Analysis / Solution

Degrees: $S_1=2, S_2=1, S_3=1, S_4=2, D_1=3, D_2=2, D_3=1$; sequence sorted $(3, 2, 2, 2, 1, 1, 1)$, sum $12 = 2 \cdot |E|$ (handshake lemma). Both drawings share $|V|=7, |E|=6$, bipartiteness and the sequence — necessary conditions. Bijection $f(S_i)=S'_i, f(D_j)=D'_j$ preserves all six edges; with equal edge counts non-edges are preserved. With reversal perms $r_S=[3, 2, 1, 0], r_D=[2, 1, 0]$, biadjacencies satisfy $A_2 = P \cdot A_1 \cdot P^T$.

7. Diagram / Graph / Truth Table / Algorithm

Edge list:

#	Edge	Clinical meaning
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1	$S_1 - D_1$	Fever indicates Influenza
2	$S_1 - D_2$	Fever indicates Measles
3	$S_2 - D_1$	Cough indicates Influenza
4	$S_3 - D_2$	Rash indicates Measles
5	$S_4 - D_1$	Fatigue indicates Influenza
6	$S_4 - D_3$	Fatigue indicates Anemia

Invariant comparison:

Invariant	Rep.1	Rep.2	Match
Vertices $ V $	7	7	✓
Edges $ E $	6	6	✓
Degree sequence	(3,2,2,2,1,1,1)	(3,2,2,2,1,1,1)	✓
Bipartite	yes	yes	✓

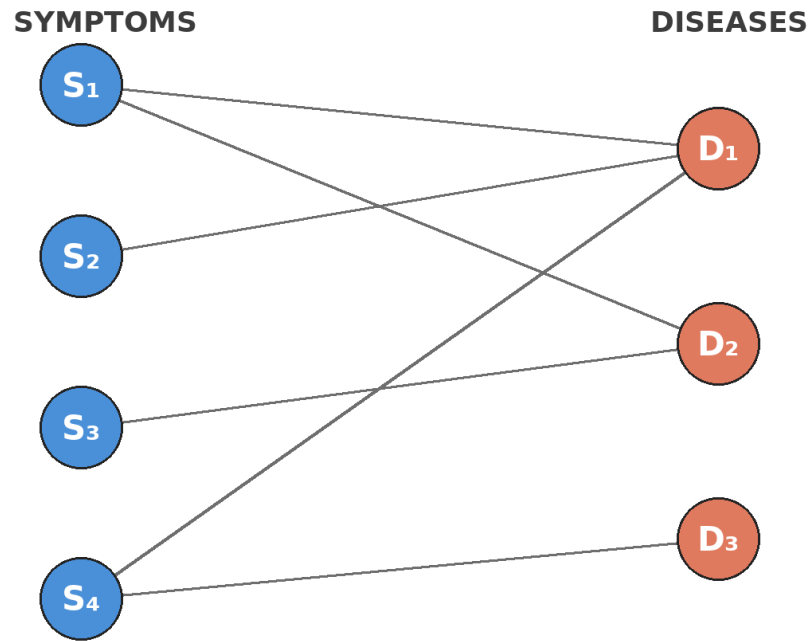


Figure 1 — Bipartite layout: symptoms left, diseases right.

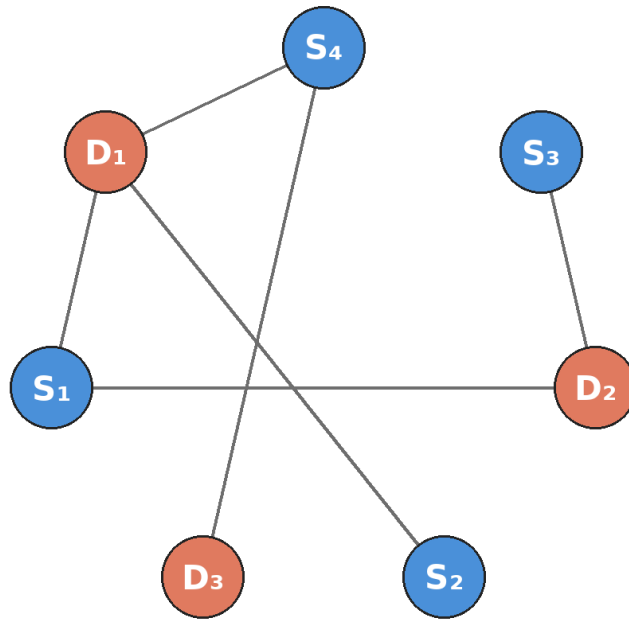


Figure 2 — Same network, circular scrambled layout. Only coordinates changed.

Biadjacency matrices:

	D₁	D₂	D₃
S ₁	1	1	0
S ₂	1	0	0
S ₃	0	1	0
S ₄	1	0	1

Reordered (S₄...S₁ | D₃...D₁):

	D₃	D₂	D₁
S ₄	1	0	1
S ₃	0	1	0
S ₂	0	0	1
S ₁	0	1	1

Algorithm — Merge check: compute A_1, A_2 , degree sequences; if invariants differ → not isomorphic; else search for P with $A_2 = P \cdot A_1 \cdot P^T$ (exhaustive for $n=7$, else use canonical labeling).

8. Interpretation of Results

Identical invariants plus the exhibited P -matrix prove the drawings are one knowledge base.

Layout is irrelevant; connectivity is the invariant. Practically, two editors can work in different tools and merge losslessly.

9. Practical Application

Deployed as the Isomorphism Explorer: one network, two layouts, toggle morphs bipartite ↔ circular with spring, clicking a vertex highlights incident edges and its degree (e.g., D_1 degree 3). Curators visualise and de-duplicate without fearing layout artefacts.

10. Innovation / Proposed Improvement

- Store the graph in a graph DB with adjacency lists; isomorphism becomes a merge tool for concurrent edits.
- Extend to a 3-partite patient-symptom-disease network for end-to-end diagnosis paths.

11. Conclusion

Bipartite modelling plus a permutation certificate turns a visual intuition (“same graph”) into a checkable artefact.

12. References

- Rosen, Chap. 10 — Graphs.
- Vite + Vitest + Playwright docs.

All counts and proofs in this report are computationally verified (see live site <https://ai-medical-diagnosis-site.vercel.app>).